16. Perimeter & Area

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Perimeter of a rectangle:

A rectangle is 7 cm long and 4 cm wide. What is its perimeter?

Answer: Add all 4 sides: 7 + 4 + 7 + 4 = **22 cm**.

Example 2 — Area of a rectangle:

A rectangle is 6 cm by 5 cm. What is its area?

Answer: Area = length × width = 6 × 5 = **30 cm²**.

Example 3 — Finding a square's side from its perimeter:

A square has a perimeter of 24 cm. What is the length of one side?

Answer: A square has 4 equal sides, so 24 ÷ 4 = **6 cm**.

Example 4 — Finding a missing side from the area:

A rectangle has an area of 28 cm² and a width of 4 cm. What is its length?

Answer: Area = length × width, so length = area ÷ width = 28 ÷ 4 = **7 cm**.

Example 5 — Multi-step: a shape with a corner removed:

A rectangular garden is 8 m by 5 m. A 2m × 2m square patio is cut out of one corner. How much garden area is left?

Answer: First find the whole garden's area: 8 × 5 = 40 m². Then subtract the patio: 2 × 2 = 4 m². Remaining: 40 − 4 = **36 m²**.

Foundation

15 questions

1. A rectangle is 6 cm by 4 cm. What is its perimeter?

2. A square has sides of 5 cm. What is its perimeter?

3. A rectangle is 7 cm by 3 cm. What is its area?

4. Which shape has a perimeter of 24cm? A) a 7cm×5cm rectangle B) a square of side 6cm C) both A and B D) neither

5. A rectangle is 9 cm long and 2 cm wide. What is its perimeter?

6. A square has a perimeter of 16 cm. What is the length of one side?

7. What is the area of a 5cm by 5cm square? A) 10cm² B) 15cm² C) 20cm² D) 25cm²

8. A square has an area of 16 cm². What is the length of one side?

9. Select ALL of these shapes that have a perimeter LESS than 20cm: A) a 3cm×2cm rectangle B) a square of side 4cm C) a 5cm×4cm rectangle D) a 6cm×6cm rectangle E) a square of side 3cm

10. A rectangle has a length of 8 cm and a width of 2 cm. What is its perimeter?

11. A rectangle is 6 cm long and 1 cm wide. What is its area?

12. Which rectangle has the larger area? A) 4cm×6cm B) 5cm×5cm C) They are equal D) Cannot tell

13. A square has sides of 7 cm. What is its perimeter?

14. A rectangle is 4 cm by 4 cm. What shape is it, and what is its area?

15. A rectangle has an area of 15 cm^2 and a width of 3 cm. What is its length?

Standard

20 questions

1. A rectangle is 8 cm by 5 cm. What is its perimeter?

2. Which rectangle has a perimeter of 20cm? A) 4cm×6cm B) 7cm×3cm C) both A and B D) neither

3. A rectangle is 7 cm by 6 cm. What is its area?

4. A square has an area of 36 cm^2. What is its perimeter?

5. A rectangle has a perimeter of 30 cm and a length of 9 cm. What is its width?

6. What is the area of a rectangle 11cm long and 2cm wide? A) 22cm^2 B) 24cm^2 C) 13cm^2 D) 20cm^2

7. A square garden has sides of 15 m. What is its perimeter?

8. Select ALL of these shapes that have an area of MORE than 25cm^2: A) 5cm×6cm B) 4cm×4cm C) 3cm×9cm D) 8cm×2cm E) 7cm×4cm

9. A rectangle is 12 cm long and 3 cm wide. What is its area?

10. Which shape has the SAME area as an 8cm×3cm rectangle? A) a 6cm×4cm rectangle B) a 24cm^2 square (side $\sqrt{24}$) C) both A and B D) neither

11. A rectangle has a width of 4 cm and its length is double the width. What is its perimeter?

12. A square has a perimeter of 28 cm. What is its area?

13. Two rectangles, 5cm×3cm and 2cm×3cm, are joined edge to edge with no overlap along their 3cm sides. What is the total area?

14. A rectangular field is 25 m by 12 m. What is its perimeter?

15. A rectangle has an area of 40 cm^2 and a length of 8 cm. What is its width?

16. A rectangle is 10 cm by 6 cm. What is its perimeter?

17. A square has an area of 49 cm^2. What is the length of one side?

18. A rectangle has a perimeter of 34 cm and a width of 7 cm. What is its length?

19. Select ALL of these shapes that have a perimeter MORE than 20cm: A) 6cm×5cm rectangle B) 4cm×4cm square C) 8cm×3cm rectangle D) 5cm×5cm square E) 9cm×2cm rectangle

20. A rectangle has an area of 54 cm^2 and a width of 6 cm. What is its length, and what is its perimeter?

Hard

5 questions

1. A gardener has a 12m by 7m rectangular lawn. He cuts a 3m×2m patio out of one corner of it. How much lawn area is left?

2. Select ALL of the shapes below that have an area of exactly 36cm^2: A) a 6cm×6cm square B) a 4cm×9cm rectangle C) a 3cm×12cm rectangle D) a 5cm×8cm rectangle

3. Two square tiles with sides of 6 cm are pushed right up against each other to form a single rectangle. What is the perimeter of the resulting rectangle?

4. A rectangle has a perimeter of 38 cm. Its length is 3 cm more than its width. What are the rectangle's length and width?

5. A rectangular room measuring 9m by 6m has a square rug of side 3m placed in the middle of the floor. What area of floor is NOT covered by the rug?

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